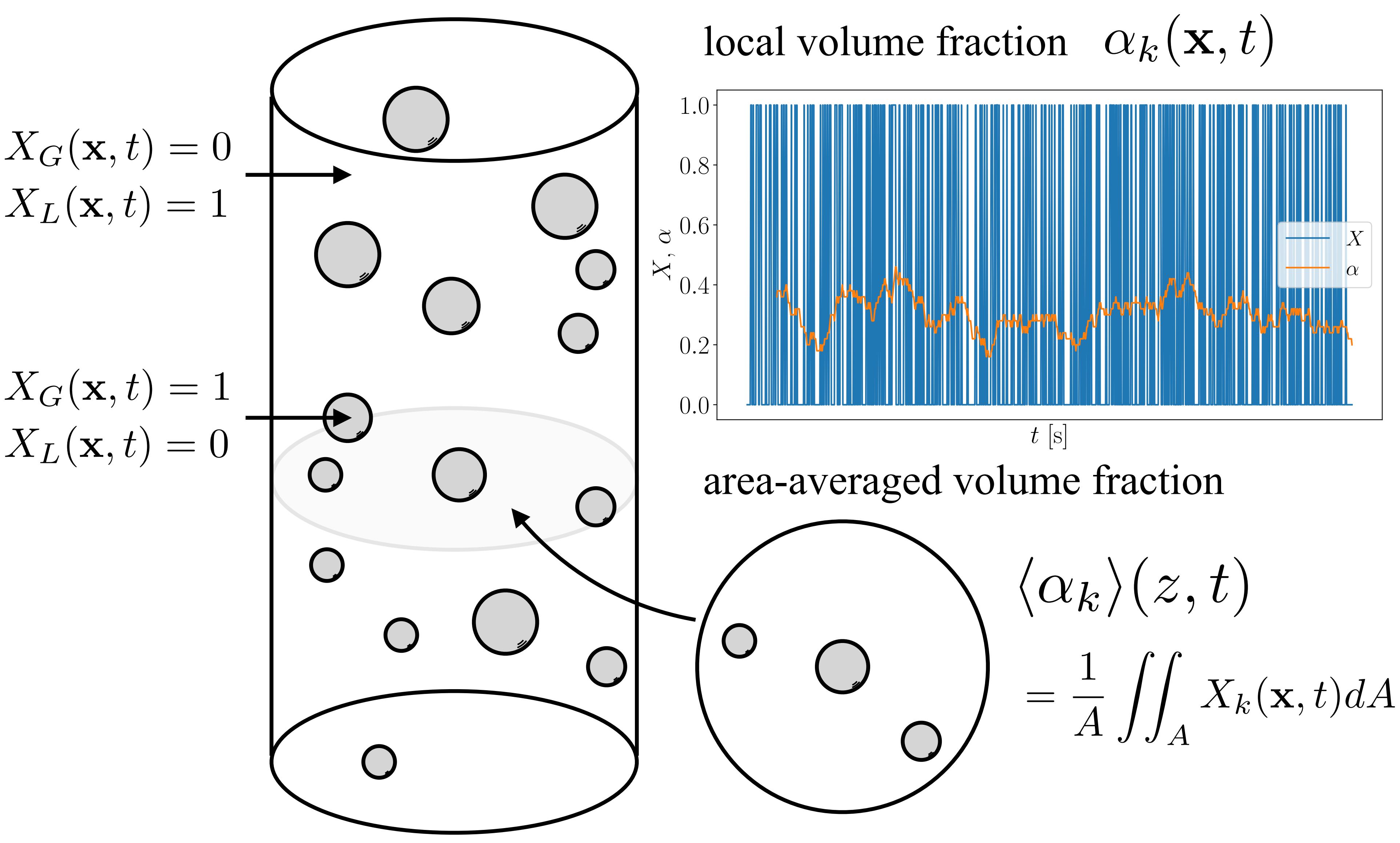
local volume fraction $\alpha_k(\mathbf{x}, t)$
$X_G(\mathbf{x}, t) = 0$
$X_L(\mathbf{x}, t) = 1$
$X_G(\mathbf{x}, t) = 1$
$X_L(\mathbf{x}, t) = 0$
1.0
0.8
0.6
0.4
0.2
0.0
X, α
X
α
t [s]
area-averaged volume fraction
$\langle \alpha_k \rangle(z, t)$
$= \frac{1}{A} \iint_A X_k(\mathbf{x}, t) dA$